

Tingjun Huang

MSc student in Robotics, Systems and Control at ETH Zurich

huangti@ethz.ch | tangenth.github.io | [GitHub: TangentH](#) | [Google Scholar](#)

Research interests: Embodied AI, 3D computer vision, robot learning and manipulation, visual navigation.

Education

ETH Zurich

MSc in Robotics, Systems and Control

Sep. 2025 - Present

Southern University of Science and Technology (SUSTech)

BEng in Information Engineering

Aug. 2021 - Jul. 2025

GPA: 3.91/4.0; rank: 2/23.

Research Experience

National University of Singapore

Research Assistant / Artificial Intelligence Research Lab

Jun. 2024 - Sep. 2024

Supervisor: Prof. Lin Shao

- Designed and implemented an autonomous social navigation agent using vision-language models for high-level task reasoning and policy planning in social environments.

Selected Research and Projects

RIGOR: Rig-Informed Geometry for Omnidirectional Reconstruction

3D reconstruction research project

Feb. 2026 - Aug. 2026

- Developed a 3D reconstruction pipeline for 360-degree construction-site walkthroughs using a frozen feed-forward model and four-view virtual camera rigs.
- Developed virtual-rig checks for local pose correction and combined cyclic four-view retrieval with joint geometric verification for loop-closure validation.

Dexterous Manipulation and PIP-DIP Coupling

Real World Robotics coursework project, ETH Zurich

Sep. 2025 - Dec. 2025

- Assembled an ORCA hand, trained and deployed a reinforcement learning policy for in-hand object reorientation, and implemented imitation learning for color-based sorting.
- Completed a team project on PIP-DIP joint coupling in the ORCA hand.

Knowledge Distillation for Visual Place Recognition

Individual research project

Sep. 2023 - Nov. 2024

- Fine-tuned DINOv2 for visual place recognition and distilled its representations into a lightweight Mamba-based model for efficient mobile deployment.

Selected Research and Projects

Visual Teach-and-Repeat Navigation

Core member

Oct. 2023 - Sep. 2024

<https://github.com/Gralerfics/pf-drive>

- Co-developed a monocular visual teach-and-repeat navigation system for differential-drive and Ackermann-steering platforms in Gazebo and Webots.

Category-Specific Object Grasping

Robotic Motion and Control coursework project

May 2024 - Jun. 2024

- Co-implemented a grasping system for the Kinova Gen3 Lite arm using GGCNN and YOLOv5 to grasp irregularly shaped objects from specified categories.

Real-Time Augmented Reality

Digital Image Processing coursework project

Apr. 2024 - Jun. 2024

<https://github.com/TangentH/AR-Gallery>

- Built an AR application combining custom ArUco marker detection with Poisson blending for seamless image fusion.

Additional Engineering Experience

SUSTech RoboMaster Team

Hardware and Embedded Systems Engineer

Sep. 2022 - Aug. 2023

- Designed custom PCBs and embedded code for mobile robots, including infrared power control and buck-boost converters; assembled and debugged components including RFID systems.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, VHDL

Robotics and development: ROS, Gazebo, Webots, Git, Linux, Bash/Zsh

Hardware: PCB design, STM32 development, SMD soldering, Keil, Vivado, EasyEDA Pro

Selected Awards

- First Prize, RoboMaster University Championship (National), China. (2023)
- Outstanding Student, SUSTech. (2023, 2024)
- Merit Student Scholarship, SUSTech (Second Class: 2022, 2023; Third Class: 2024).
- Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), China. (2022)

Languages

Mandarin and Cantonese: Native | English: TOEFL 107 | German: Beginner